Cayleb Alvarez-James

AI Agent Systems Engineer - Autonomous Agents, Agent Harnesses & Self-Improving Infrastructure

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Summary

AI agent systems engineer and technical founder. I have spent the past year on agentic systems and harness engineering, starting while I was finishing high school, where I was also wrestling team captain for two years. For most of that stretch I worked 13 to 14 hours a day on it, around school, my job and wrestling. I work across several harnesses instead of just one: OpenCode and DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex and Claude Code. I pick whichever has usage left and fits the job. Right now most of the work runs through Muse, Claude Duet and the DSH Fleet I am finishing. Three systems make up most of it. Ergon's employee agents set themselves up: the agent watches how someone works, asks them targeted questions, learns the role and keeps a memory of the person, adding skills and tools as it goes. The DSH Fleet is a distributed orchestration layer: it runs dynamic swarms across desktop, virtual machine and browser surfaces, lets outside agents join through Slack, and improves its own workflows, skills and plugin behaviour from the results of finished jobs. OpenBell is my trading research system: I set the weights of tiny models directly, freeze their predictions before scoring, and score them in time order with costs counted. 1,200+ GitHub contributions in the last year. 4,000+ automated tests across five shipped systems.

Experience

Founder - Independent AI Agent Systems | Oct 2025-present

The current builds:

Selected Projects

Ergon - Self-Configuring Employee Agents & Autonomous Role Learning

Node.js, C#, Python, SQLite

Ergon case study

- Once installed, the agent watches recurring work and tools, asks targeted questions, and builds a persistent model of the role. It proposes and builds a capability, then puts it through evaluation and review before earning the right to use it.
- Proven skills enter a shared library, earnable through the same gate; permissions stay reversible.
- Standalone, in development: signed Windows installer and taskbar app, demo connectors, ticket-scoped remote repair. RHL: a proposed first deployment - planned, not implemented. 622 tests pass.

DSH Fleet - Distributed, Self-Improving Agent Infrastructure

Python, TypeScript

DSH Fleet case study

- Control plane spanning desktop, Windows VM, phone VM and browser surfaces. Slack is the open coordination layer: my agents and bring-your-own external agents share one ticket, context and outcome protocol. A lead forms a task-shaped swarm per objective - no standing bench - dissolving after one auditable outcome.
- Each objective gets a dynamic swarm shaped to the task, formed around a lead: preferences, current skills, research, model routing. Every session is defined, material actions need approval, and the swarm dissolves after one recorded outcome.
- Job evidence feeds Omni Loop revisions to workflows, plugin behaviour, the skill library and the loop itself. Promotion needs a fresh check measuring a win on a fixed scorecard (checker-pass rate, cost per checker-passed task) plus a frozen regression set; failures archived, changes reversible. Inside every swarm: preferences first, current skill second, raw model knowledge last. The fleet has caught its own swarms filing false 'done' reports - each found by re-checking the primary source.

OpenBell - Weight-Space Training & Evaluation for Trading Models

Python, Rust

OpenBell case study

- Direct weight-space construction: low-dimensional weights for tiny specialist models, not prompted strategies. Weights and predictions frozen before scoring: inspectable, falsifiable.
- Adversarial evaluation: chronological splits, transaction costs, negative results retained, pre-registered kill rules, no money test until training clears the exit-geometry breakeven. Distillation failed on both benchmark instruments; directly authored weights produced the strongest results - one Nasdaq configuration net-positive on historical data.
- Kalshi, futures and meme desks: live, capital-isolated proving grounds - parallel slots, 2x ladder runs, reconciled accounting, small real capital, pre-registered kill conditions.

Sophos - Coding-Agent Desktop App

React, TypeScript, Tauri

github.com/cayleb-james2008/sophos

- Coding-agent desktop app; sessions and transcripts survive restarts.
- 1,224 interface tests, 82 Node tests and 56/56 Windows checks pass; merged after full hosted CI.

dotz - AI Coding Agent Dashboard

Rust, Tauri, JavaScript, SQLite

github.com/cayleb-james2008/dotz

- Desktop app running and supervising AI coding agents: model choice, safe process-tree cleanup.
- 828+ core tests pass on each of Windows, macOS and Linux; all 6 hosted check jobs pass; merged.

Asmodeus - Trading Research on Past Data

Rust

Asmodeus case study

- Tests trading ideas on historical data in strict time order, with fees and safeguards against peeking ahead; rejected ideas logged. 1,111 Rust tests pass. Research only.

Industry AI Suite - Source-Checked Review Tools

Python

github.com/cayleb-james2008/industry-ai-suite | Review history implementation

- Ten review tools; every finding shows its source and a person decides. 382 Python tests pass.

Skills

AI & agents: Multi-harness agent operations (OpenCode/DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex, Claude Code), autonomous role learning, dynamic-swarm orchestration, recursive self-improvement systems (the Omni Loop), evaluation design, earned permissions and approval gates

Languages: Python, TypeScript, JavaScript, Rust, HTML/CSS

Models & data: Weight-space model construction, linear/LDA specialists, chronological backtesting, experiment design, React, Tauri, SQLite, web APIs, Git/GitHub, Linux

Additional Experience

Robert H. Lord Company (RHL) - AI Concept Discussion & Warehouse-Board Demo | Sep 2026-present

David Cooke Plaster Co. - Plaster Work | Summer 2026

Dunkin' | 2024-2026

Tommy's Pizza | Summer 2024

Education

Portland High School | Portland, Connecticut | Graduated 2026

Coursework: Cybersecurity; AP Computer Science